IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26j-437

1. Topic (12 words max)

Integration of a Modular, Emotionally-Aware, Blockchain-Verified, Fault-Tolerant Assistive Robot

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

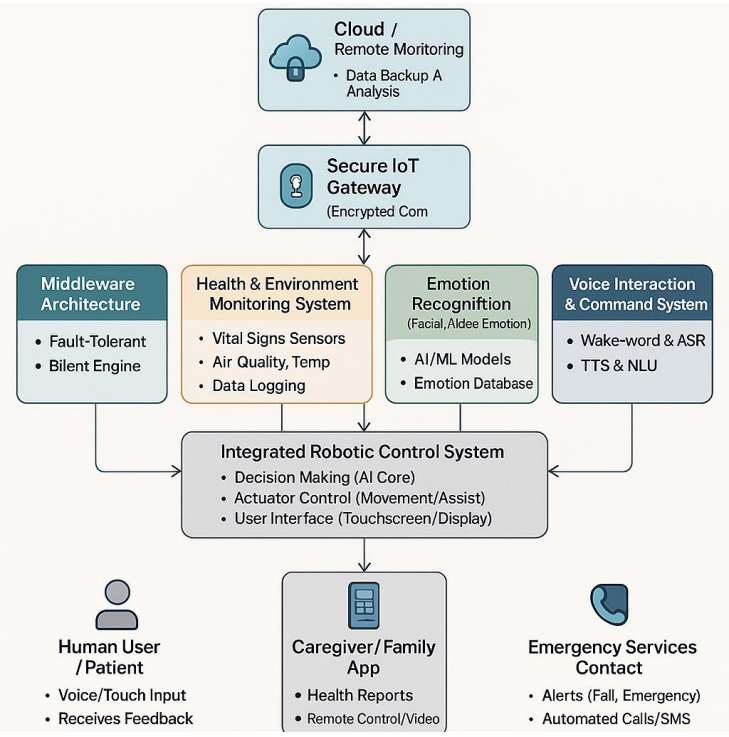
Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The development of assistive robots that can interact safely, intelligently, and empathetically with humans remains a key challenge in modern robotics. Current systems often lack essential capabilities such as real-time adaptability, fault tolerance, secure skill integration, and emotional awareness—critical features for effective operation in domestic environments. While platforms like ROS2 provide modular flexibility, they do not inherently support runtime recovery from failures, intent-based behavior coordination, emotion recognition, or blockchain-backed trust mechanisms. This research addresses these gaps by integrating four core components—fault-tolerant middleware, real-time environmental and health monitoring, emotion recognition, and voice-based command processing—into a unified robotic architecture. The resulting system, inspired by Baymax, enables self-repair, adaptive emotional responses, secure plugin-based skill updates, and context-aware decision-making, offering a novel student-led implementation of an advanced, industrial-grade assistive robot.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution integrates five key research modules into a cohesive robotic architecture built on ROS2, designed to enhance modularity, fault tolerance, and emotional intelligence. At its core is the Middleware Layer , which functions as the central nervous system, orchestrating interactions between subsystems, managing faults, and enabling plugin-based behavior injection. The Intent Engine prioritizes and interprets multi-modal inputs—such as emotion, voice, and environmental data—to determine the robot's real-time actions. A Sensor Fusion System combines readings from gas, temperature, humidity, and vital sign sensors to ensure both environmental and user safety. The Emotion Recognition Module analyzes facial expressions and voice tone to detect user emotions and influence the robot's responses accordingly. Meanwhile, the Voice Interaction System supports natural language commands, contextual understanding, and emotionally adaptive dialogue. All operations are securely logged using a blockchain layer , ensuring transparency and traceability, while also supporting secure, verifiable plugin-based skill upgrades. A conceptual diagram would illustrate these components interconnected via an event bus, feeding into the Intent Engine and Middleware for coordinated, intelligent behavior. This comprehensive, student-led integration represents a novel advancement in academic robotics, combining industrial-grade features like self-repair, emotional awareness, and secure capability evolution into a single, functional assistive robot platform.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

1. Domain Expertise

This project requires cross-disciplinary expertise in robotics, AI, and secure systems. A strong foundation in ROS2 architecture is essential for building a modular, event-driven, and fault-tolerant middleware. Specialists must implement self-repair mechanisms, such as node auto-restart, failure detection, and state persistence, to maintain operational continuity. Knowledge in blockchain frameworks like Hyperledger Fabric is needed to ensure secure action logging and plugin verification. In the emotional intelligence domain, expertise in machine learning is crucial for facial and voice emotion recognition using datasets like FER2013, CK+, and RAVDESS. NLP proficiency is also required to build emotionally adaptive dialogue systems using transformer models (e.g., BERT) and libraries like spaCy and NLTK. Lastly, skills in multi-sensor data fusion, particularly Kalman filtering, are necessary to synchronize and interpret data from environmental and physiological sensors, ensuring coherent sensory input and adaptive robot behavior.

2. Data Requirements

The system requires diverse, high-quality multimodal datasets to support emotionally intelligent and context-aware robotic behavior. For **facial emotion recognition**, datasets like FER2013 and CK+ provide labeled facial expressions. **Voice emotion analysis** leverages audio datasets such as RAVDESS, containing speech and song recordings with emotional annotations. For **natural language processing**, emotion-tagged conversational datasets like EmotionLines and intent classification corpora like CLINC150 are employed to train adaptive dialogue systems. On the sensory side, environmental and physiological data—including temperature, gas, heart rate, and SpO_2—can be sourced or simulated from datasets such as PhysioNet for vitals and environmental sensor benchmarks from the UCI ML Repository. **Kalman filtering** is applied for synchronizing noisy real-time inputs. Additionally, blockchain-based metadata (e.g., plugin logs and execution traces) is structured for auditability, enabling system transparency, resilience, and trust in autonomous decision-making.

Dataset Links:

- FER2013: https://www.kaggle.com/datasets/msambare/fer2013
- CK+ (Extended Cohn-Kanade): https://www.jeffcohn.net/resources/
- RAVDESS: https://zenodo.org/record/1188976
- EmotionLines: https://github.com/yangheng95/EmotionX
- CLINC150: https://github.com/clinc/oos-eval
- PhysioNet: https://physionet.org/
- UCI ML Repository (Sensor data): https://archive.ics.uci.edu/

8. Objectives and Novelty

Main Objective To develop an integrated, real-time assistive robot that combines emotion-aware interaction, adaptive navigation, health and hazard monitoring, and self-repair capabilities within a secure, modular ROS2-based architecture.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Praveen S.K.S IT22114044	Enables command-based navigation using voice recognition and intent extraction for guided robot movement	• Integrate RPLIDAR and onboard sensors with ROS2 nodes • Implement SLAM for real-time mapping and localization • Develop obstacle detection and sensor fusion logic • Design path planning with Active SLAM and RL-based adaptation • Enable voice-command-based guided navigation • Test mobility framework in dynamic indoor environments	Most use **static path plans**; we will use **Active SLAM** that adapts to dynamic changes. We'll incorporate **voice-guided movement** (e.g., "Come here").Use of lightweight RL (reinforcement learning) in real-world path optimization is rare.
S.N.S.B.K.K.Semasinghe IT22084668	Monitors health and environmental hazards using real-time sensor fusion for proactive	• Integrate and interface sensors (DHT11, MQ-2, MPU6050, MAX30100) with ROS2 nodes • Develop real-time sensor fusion algorithms to combine	Most systems do **only vital signs** or **only environmental data**. We're doing **sensor fusion** to combine both — and take automated emergency

	emergency response.	environmental and physiological data • Implement event detection logic for emergencies (falls, fevers, gas leaks) • Set up alert notification system via Firebase and Twilio APIs • Design a unified ROS2 framework to manage multi-sensor data and triggers • Test and validate system responsiveness in simulated and real-world scenarios	action. All data flows into our ROS2 system in real time.
Madushanka L.A.S IT22602114	Detects user emotions via facial and voice analysis to trigger empathetic speech and gestures.	• Integrate camera and microphone input with ROS2 nodes on Raspberry Pi • Develop facial emotion recognition pipeline using OpenCV and TensorFlow with FER2013 • Implement voice emotion analysis using pre-trained models on RAVDESS dataset • Design emotion classification and fusion logic for multi-modal input • Integrate chatbot API (e.g., Dialogflow) for empathetic speech response generation • Program gesture control system to trigger context-appropriate robot movements	Existing systems typically use **either** facial recognition **or** speech sentiment. We will implement **real-time multi-modal emotion fusion** (face + voice) **and** generate natural, empathetic responses via speech + gestures. Few current systems deploy this in a real-time assistive robot.

		• Test and optimize real-time performance in realistic interaction scenarios	
Mallawaarachchi U.V.N IT22092274	Enables fault recovery, intent arbitration, secure plugin updates, and blockchain-based activity logging.	• Design and implement fault-tolerant ROS2 middleware for module supervision and auto-recovery • Develop secure plugin management system enabling dynamic skill updates • Implement intent prioritization and arbitration mechanisms for user command handling • Integrate blockchain (e.g., Hyperledger Fabric) for immutable logging of actions and plugin verification • Create APIs for middleware communication and plugin interfacing • Test middleware robustness, security, and scalability in simulated and real-world scenarios	ROS2-based middleware that combines fault-tolerance, secure extensibility, and context-aware arbitration into a plug-and-play framework for assistive robots.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Praveen S.K.S IT22114044	This project aligns with software engineering specialization by applying modular ROS2-based system design, real-time sensor integration, and algorithm development. It involves implementing Active SLAM, sensor fusion, and lightweight reinforcement learning for adaptive path planning—core software engineering skills in embedded and robotics software. The integration of voice-command processing and fault-tolerant middleware showcases expertise in event-driven architectures and human-computer interaction. Using platforms like Arduino and Jetson Nano emphasizes cross-platform development and hardware-software interfacing, crucial in software engineering for robotics. Overall, this research demonstrates designing, developing, and testing complex, scalable, and maintainable software systems in dynamic real-world environments.
S.N.S.B.K.K.Semasinghe IT22084668	This project exemplifies software engineering specialization by developing a robust, real-time multi-sensor data fusion system within a modular ROS2 framework. It involves integrating diverse hardware sensors and designing efficient algorithms to process physiological and environmental data concurrently. The implementation of alert mechanisms using cloud services like Firebase and Twilio highlights skills in API integration and event-driven software design. By combining health and hazard monitoring into a unified platform, this research demonstrates expertise in building scalable, fault-tolerant, and maintainable software systems tailored for assistive robotics in real-world settings.
Madushanka L.A.S IT22602114	This project aligns with software engineering by integrating computer vision, audio processing, and natural language processing into a cohesive, real-time system. It demonstrates expertise in deploying machine learning models using frameworks like TensorFlow and OpenCV, alongside API integration for chatbot-driven empathetic dialogue. The modular design on Raspberry Pi reflects skills in embedded system programming and resource optimization. Overall, it showcases building scalable, responsive software that bridges human emotions and robotic interaction, highlighting core software engineering principles applied to assistive robotics.
Mallawaarachhi U.V.N IT22092274	This project demonstrates advanced software engineering skills by designing a fault-tolerant, modular middleware architecture using ROS2. It integrates secure plugin management for dynamic skill updates and implements intent-driven arbitration to prioritize user commands effectively. The incorporation of blockchain technology for immutable logging highlights expertise in secure, decentralized system design. Overall, it reflects the development of scalable, maintainable middleware that enhances reliability, security, and intelligence in assistive robotics—core competencies within the software engineering discipline.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms	Hansi	De Silva	
Co-Supervisor	Mr	Ashvinda	Iddamalgoda	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes	✓	No	

b) Does the proposed topic exhibit novelty?

Yes	✓	No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes	✓	No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes	✓	No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable as a final year research topic.
check the data availability.

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.